ALESSIO RUSSO

Research Fellow

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WORK EXPERIENCE

Research Fellow

University of Parma @ Vitality Lab

Feb 2026 – Oct 2026 Parma, Italy

- Investigated the recognition and synthesis of expressive human motion from motion-capture data, with applications to Human–Robot Interaction.
- Developed Deep Learning pipelines for classifying and synthesizing wrist motion profiles using discriminative and conditional generative models.
- Evaluated synthetic movements through cross-domain classification, distributional analysis, and a human perceptual study.

Research Fellow

University of Parma @ CyberSec Lab

Dec 2024 – Sep 2025 Parma, Italy

- Contributed to the *SERICS* (Security and Rights in the CyberSpace) project on Deep Learning-based cybersecurity applications.
- Focused on identifying exploitable software vulnerabilities in C++ codebases.
- Developed automated vulnerability classification pipelines using LLM fine-tuning and in-context learning.

Software Developer

REM Software & Automation Srl

Jul 2022 – Mar 2024 Modena, Italy

- Developed *Machine Assistant*, an industrial voice-interaction system.
- Designed and implemented Python backend components for voice recognition pipelines.
- Designed and implemented a smartwatch application in Kotlin.
- Integrated C++-based industrial machinery control systems to enable voice commands.

EDUCATION

Master's Degree in Computer Science

University of Parma

Sep 2024 – Jul 2026

Final Grade: 110L/110

Master's Thesis: *From Recognizing Vitality Forms-Driven Human Actions to Synthesizing Expressive Motion Profiles via Deep Learning*

Bachelor's Degree in Computer Science

University of Parma

Sep 2020 – Jul 2024

Final Grade: 101/110

Bachelor's Thesis: *Design of an Artificial Intelligence System for Malware Detection and Classification.*

PROJECTS

Fenn (Friendly Environment for Neural Networks)

- Open-source Python framework for managing and structuring Machine Learning experiments.
- Automates configuration handling, logging, experiment tracking, and notifications.
- Supports integration with tools such as TensorBoard and Weights & Biases.
- Maintained by a community of 50+ contributors, with 80+ GitHub stars and 70+ forks.

Code: https://github.com/pyfenn/fenn

Reinforcement Learning-Based Agent Design

- Implemented Reinforcement Learning agents using *Gymnasium* environments.
- Designed neural policy models based on *MLP* and *CNN* architectures.
- Trained policies in control and arcade-style environments using Deep Reinforcement Learning methods.

Code: https://github.com/blkdmr/rl-algorithms

Quality Assessment

- Computer vision pipeline for automated potato quality inspection.
- Object detection using YOLO to localize potatoes in images.
- Image classification via fine-tuned ResNet-50 to distinguish good and defective samples.

Code: https://github.com/blkdmr/potatoes

PUBLICATIONS

- Alessio Russo, Eleonora Iotti, Alessandro Dal Palù. *Answer Set Programming for Feature-Based Explanation of Malware Prediction*. In: *Proceedings of the 40th Italian Conference on Computational Logic (CILC 2025)*, CEUR Workshop Proceedings, vol. 4003, 2025.
- Alessio Russo, Alessio Saccuti, Eleonora Iotti, Giuseppe Di Cesare. *Analysis and Generation of Motion Profiles from Positive and Negative Human Actions*. In: *Frontiers in Artificial Intelligence, 9, 1963524, 2026.*

SKILLS

- **ML & Scientific Computing**: PyTorch, scikit-learn, NumPy, pandas, SciPy, Matplotlib, OpenCV, Gymnasium
- **DevOps**: Git, TFS, Docker, Podman, Linux, Bash
- **Databases**: SQL, PostgreSQL, Redis, Firebase

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